PBL Project Evaluation Phase-1

Machine Learning – 22CSE61

VI Sem

The Department of Computer Science & Engineering

LegalEase:Smart Contract Intelligence System Using NLP for Risk Detection

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Real World Scenario Example

Rahul signs a gym membership contract

Hidden clauses:

- Auto renewal every year
- 60-day cancellation rule
- Late cancellation = full year fees

Rahul didn't understand the contract → loses ₹12,000.

Problem:
Most users sign contracts without understanding risks.

Introduction

Legal contracts are written in complex legal language called **“legalese.”**

Many users sign agreements without fully understanding the terms and risks.

LegalEase is an AI-powered system that:

- Analyzes legal contracts
- Detects risky clauses
- Explains them in simple language

Technologies used:

- Natural Language Processing (NLP)
- Transformer models
- Retrieval-Augmented Generation (RAG)

Problem Statement

Contracts such as rental agreements, loan documents, and online Terms of Service are difficult for non-experts to understand.

Reasons include:

- Complex legal terminology
- Hidden conditions and penalties
- Long documents that users rarely read fully

As a result, users unknowingly agree to:

- Hidden fees
- Automatic renewals
- Strict cancellation rules
- Privacy waivers

Existing tools mainly **summarize documents** but rarely detect **specific legal risks.**

Objectives

1. **Automated Risk Detection**

Identify high-risk and predatory clauses using NLP and semantic analysis.

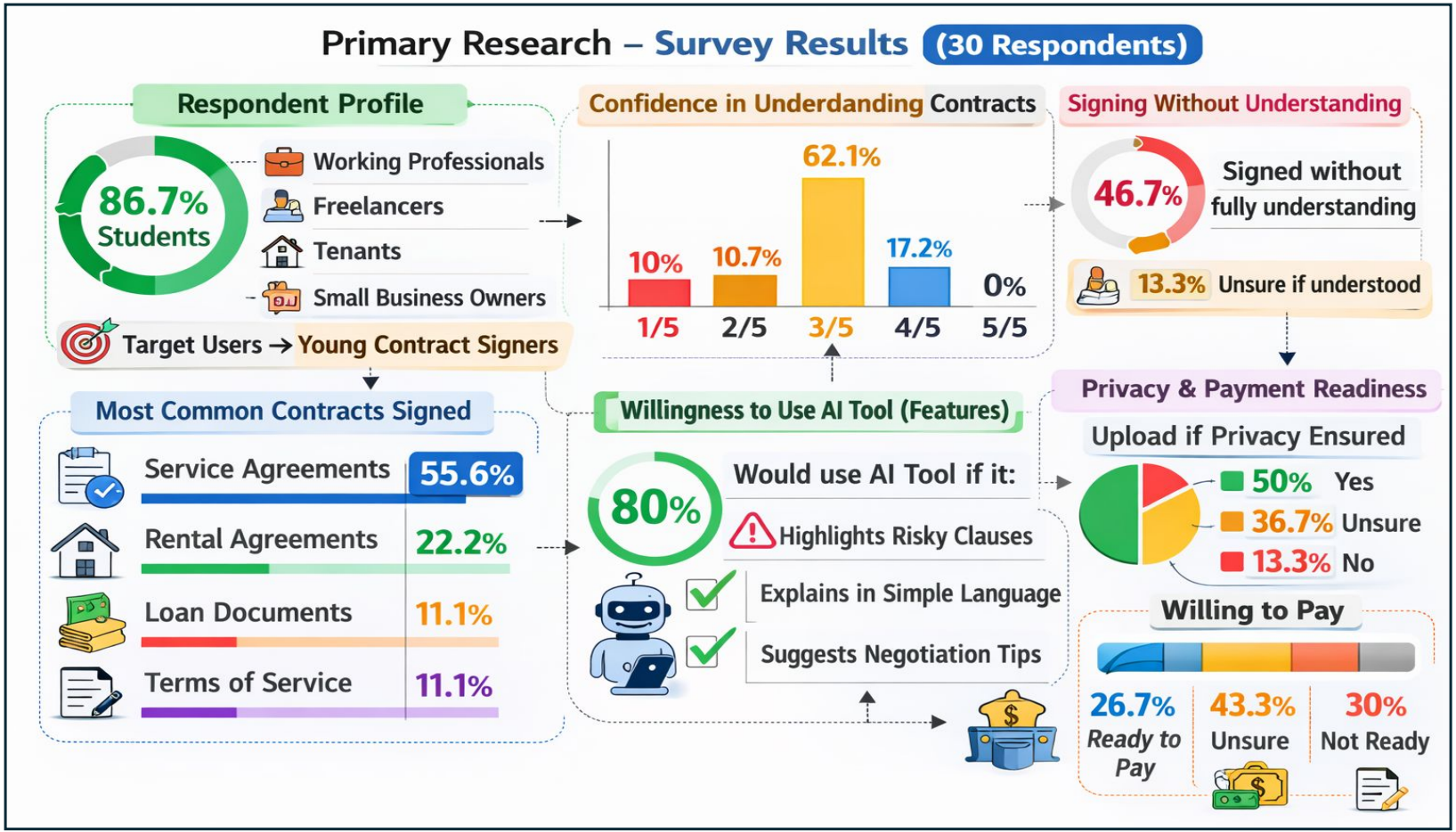
2. **Plain English Interpretation**

Convert complex legal language into simple, understandable explanations.

3. **Knowledge-Augmented Analysis (RAG)**

Compare contract clauses with a legal knowledge base for accurate risk assessment.

Summary of Primary Research



Summary of Primary Research

Analysis of Findings

•Legal literacy gap clearly exists.
•Majority have moderate or low confidence.
•Nearly half sign contracts without understanding.
•Strong demand (80%) for intelligent AI risk detection.
•Privacy assurance is critical for adoption.
•Users want clause-level insights, not just summaries.
•Market shows willingness if value is proven.

Summary of Primary Research

•Primary research validates that:
•Users struggle to understand legal contracts.
•Hidden risks and complex language are key barriers.
•There is strong acceptance of AI-assisted risk detection.
•Privacy-focused, clause-level analysis like **LegalEase** has strong real-world demand.

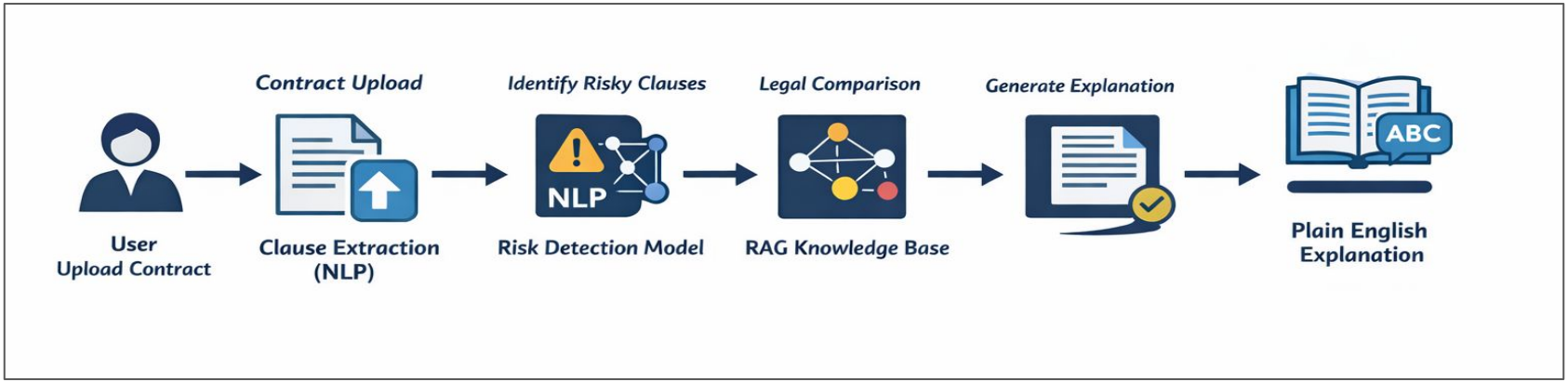
Summary of Secondary Research

- Studies show transformer-based models (Legal-BERT, GPT-4) achieve high accuracy (~83%) in clause classification — validating LegalEase's automated risk detection module.
- NLP reduces contract review time by up to 76%, proving AI can efficiently analyze lengthy rental, service, and internship agreements.
- Clause extraction and risk prediction are technically feasible using supervised learning — supporting LegalEase's high-risk clause identification feature.
- Semantic similarity techniques allow comparison of clauses with known unfair practices — directly supporting LegalEase's RAG-based knowledge comparison system.
- AI-assisted review can match expert-level quality when properly designed — strengthening the credibility of LegalEase.

Summary of Secondary Research

- Manual contract review is costly and prone to oversight, especially for non-legal users — confirming the need for a consumer-focused AI assistant.
- Data privacy and confidentiality are major concerns in legal AI — validating LegalEase's privacy-assured architecture.
- Lack of interpretability reduces trust — justifying LegalEase's plain-English explanation feature.
- Over-reliance on automation can create risks — supporting LegalEase's assistive (not replacement) approach.
- Research recommends a human-in-the-loop model — aligning with LegalEase as a decision-support tool rather than a legal authority.

WorkFlow of our Project



Key Features of LegalEase:

- Contract upload support (PDF, DOCX, images)
- OCR for scanned documents
- Camera-based contract scanning
- Clause detection using NLP
- Risk highlighting system
- Plain English explanations
- Audio explanations (Text-to-Speech)
- Multilingual explanation support

Business Model Canvas

Key Partners
- LLM Providers
- Legal Experts
- Freelance & Rental Platforms
- Consumer Protection Groups

Key Activities
- Model Training & Optimization
- Knowledge Base Updates
- Platform Development
- Security & Compliance

Key Resources
- LLMs (GPT-4 / Llama 3)
- Legal Knowledge Base (RAG)
- Vector Database
- AI/ML Team
- Cloud Infrastructure

Value Propositions
- AI-powered contract risk detection
- Plain English explanations
- Contract risk score system
- RAG-based legal comparison
- Affordable & privacy-safe solution

Customer Relationships
- Self-service SaaS
- Freemium Model
- Subscription Plans
- AI Chat Support

Channels
- Web App (Primary)
- Freelance & Rental Platform Integrations
- Legal Aid Partnerships
- Digital Marketing

Customer Segments
- Freelancers
- Tenants & renters
- Gig workers
- Small business owners
- Startup founders
- Individuals signing loans or service agreements
- Consumer rights organizations
- Legal aid clinics

Cost Structure
- Cloud & API Costs
- Development Team
- Legal Consultants
- Marketing & Compliance

Revenue Streams
- Monthly/Annual Subscriptions
- Pay-per-Contract
- Premium Risk Reports
- B2B API Licensing

Clause Trap Chain Detection

Some clauses appear harmless individually but become dangerous when combined.

Example:

Clause 3 — Subscription renews automatically
Clause 7 — Cancellation requires 60-day notice
Clause 12 — Late cancellation requires full payment
Clause 15 — Disputes resolved through arbitration

Together these clauses create a **Renewal Lock-In Trap.**

LegalEase detects such hidden clause combinations.

Clause Trap Chain Detection

Risk Analysis Output

Trap Chain Detected: Clause 3 → Clause 7 → Clause 12 → Clause 15

Risk Type: **Renewal Lock-In Trap**

Severity: 🔴 High Risk

Explanation:Users may unknowingly become locked into automatic yearly payments with limited ability to cancel.

LegalEase alerts users before signing the contract.

Innovation and Impact

LegalEase goes beyond simple document summarization.

Key contributions:

- Clause-level risk detection
- Knowledge-based risk comparison using RAG
- Plain language explanations for non-experts
- Accessibility through audio explanations

Impact:

- Improves legal awareness
- Reduces risk of unfair agreements
- Supports safer digital transactions

Future Scope

Future improvements include:

- Mobile application support
- Expansion of legal clause knowledge base
- Real-time contract scanning
- Integration with digital signing platforms
- More advanced AI models for legal reasoning

LegalEase can evolve into a **consumer protection AI assistant.**